

Using Time Series Analysis for Sales and Demand Forecasting

Stakeholder Report

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FourthRev

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1. Problem Overview

Nielsen is seeking to invest in developing a new service aimed at small to medium-sized independent publishers. This service is aimed at supporting publishers in using historical sales data to make data-driven decisions about their future investment in new publications. Their publisher customers are interested in being able to make more accurate predictions of the overall sales profile post-publication for better stock control and initial investment, but they are also interested in understanding the useful economic life span that a title may have.

The dataset is separated into two files: One file is an **ISBN list** that contains industry-standard metadata about each book, and the other is a **UK weekly trended timeline** that contains weekly sales data. To aid a more comprehensive prediction approach, appropriate classical, machine learning, deep learning and hybrid models were trained and tuned to identify sales patterns that demonstrate seasonal trends. This information would be applied Nielsen to their commission and print volume strategy to be more successful in acquiring titles that have longevity.

2. Overview of Approach

The methodology combined statistical time-series models, ensemble gradient boosting, sequential neural networks, and hybrid architectures. (**Table 1**).

Model	Type	Parameters included	Key Information
SARIMA	Statistical time-series	order (p,d,q), seasonal_order (P,D,Q,s)	Autoregressive integrated moving average with seasonal decomposition. Captures trend, seasonality, and residual structure. Selected via Auto-ARIMA using AIC.
XGBoost	Ensemble ML	learning_rate, max_depth, n_estimators, subsample, colsample_bytree, reg_alpha, reg_lambda	Gradient boosting algorithm building decision trees sequentially. Supports regularization to prevent overfitting. Recursive multi-step forecasting for 32-week horizon.
LSTM	Deep learning (sequential)	num_lstm_layers, units, dropout_rate, optimizer, learning_rate, epochs	Long short-term memory network learning temporal dependencies through gated memory cells. Trained via KerasTuner with early stopping.
Hybrid (Sequential)	SARIMA + LSTM	SARIMA residuals fed into LSTM; LSTM forecasts residuals	SARIMA captures structural trend-seasonality; LSTM models nonlinear residual patterns. Final prediction = SARIMA

			forecast + LSTM residual forecast.
Hybrid (Parallel)	SARIMA + LSTM (weighted)	sarima_weight, lstm_weight (optimized via grid search)	SARIMA and LSTM trained independently on raw data. Final prediction = weighted average of component forecasts.

Table 1: Classical, Machine & Deep Learning techniques and Interpretation methods applied.

2.1. Time Series Analysis Workflow

The end-to-end process used for identifying key sales patterns that exhibit clear seasonal trends or other distinctive characteristics is visualized below (**Figure 1**). This included preprocessing the raw data, including filtering the ISBNs and preprocessing missing values. Two books were chosen for further analysis: The Very Hungry Caterpillar and The Alchemist.

The series for both books was decomposed and ACF and PACF analysis was carried out on the volume/sales data. Further methods including ADF & Ljung Test for stationarity check and auto ARIMA to derive SARIMA and SARIMAX, were used.

Further techniques used included Machine and Deep Learning models (XGBoost and LSTM) and hybrid models (SARIMA-LSTM sequential and parallel combinations). Weekly and monthly data series XGBoost and auto ARIMA models for both books were derived for further comparison.

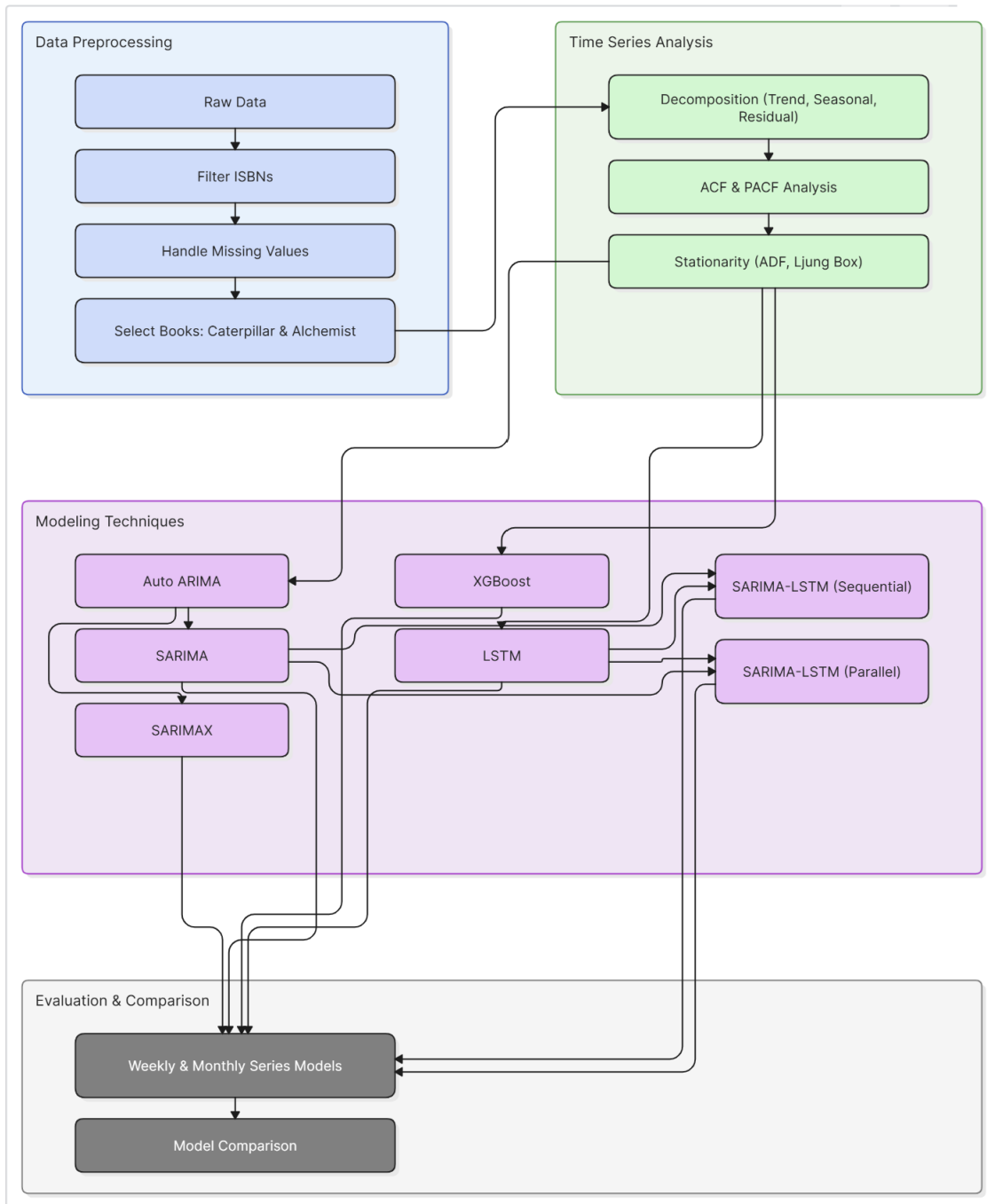


Figure 1: A high-level visualization of the time series analysis pipeline.

2.2. Data and Features

The ISBN list data set which contained several tabs, each representing the top-selling books for that product category and the UK weekly trended timeline data, were each turned into a single data frame. The UK weekly data reflected a mixture of qualitative and quantitative weekly data including 227,224 rows and 14 columns. The ISBN data reflected a mixture of qualitative and quantitative weekly data including 500 rows and 11 columns.

The visualization of the data reflects that majority of the uniquely identified books, the number of books sold or distributed during the first few years drastically declined after the first 5 years with many flatlining after.

This trend could reflect the shift to digital media and a change in reading habits (**Figure 2**).

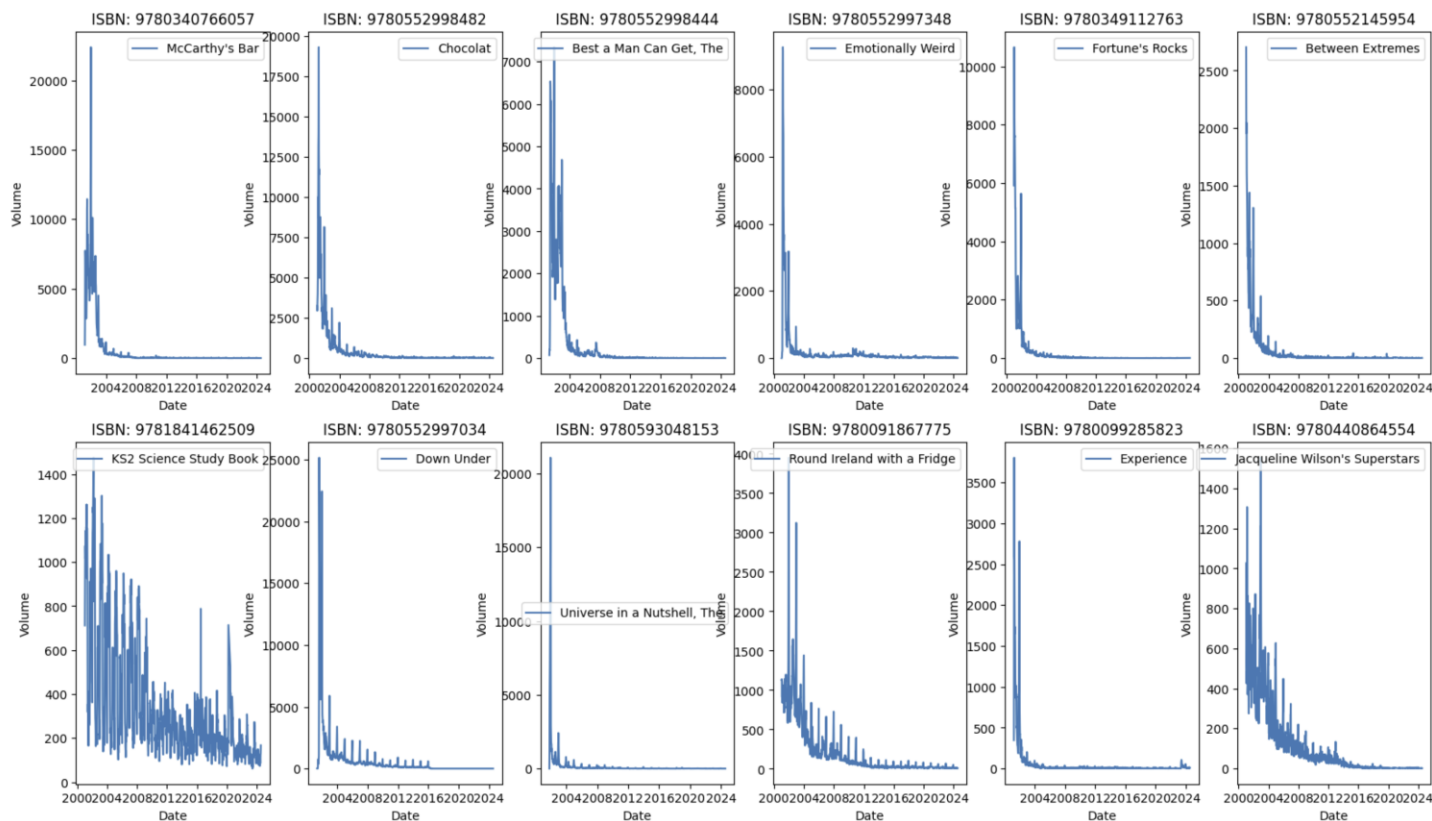


Figure 2: Visualization of Nielsen ISBN list volume/sales trend.

Further trend analysis was carried out in two focus books. The sales data for both books reflect some trend components and seasonal spikes over time. The Very Hungry Caterpillar reflects seasonal spikes which clearly rise over time, and The Alchemist reflects peak height increase over the years (**Figure 3**).

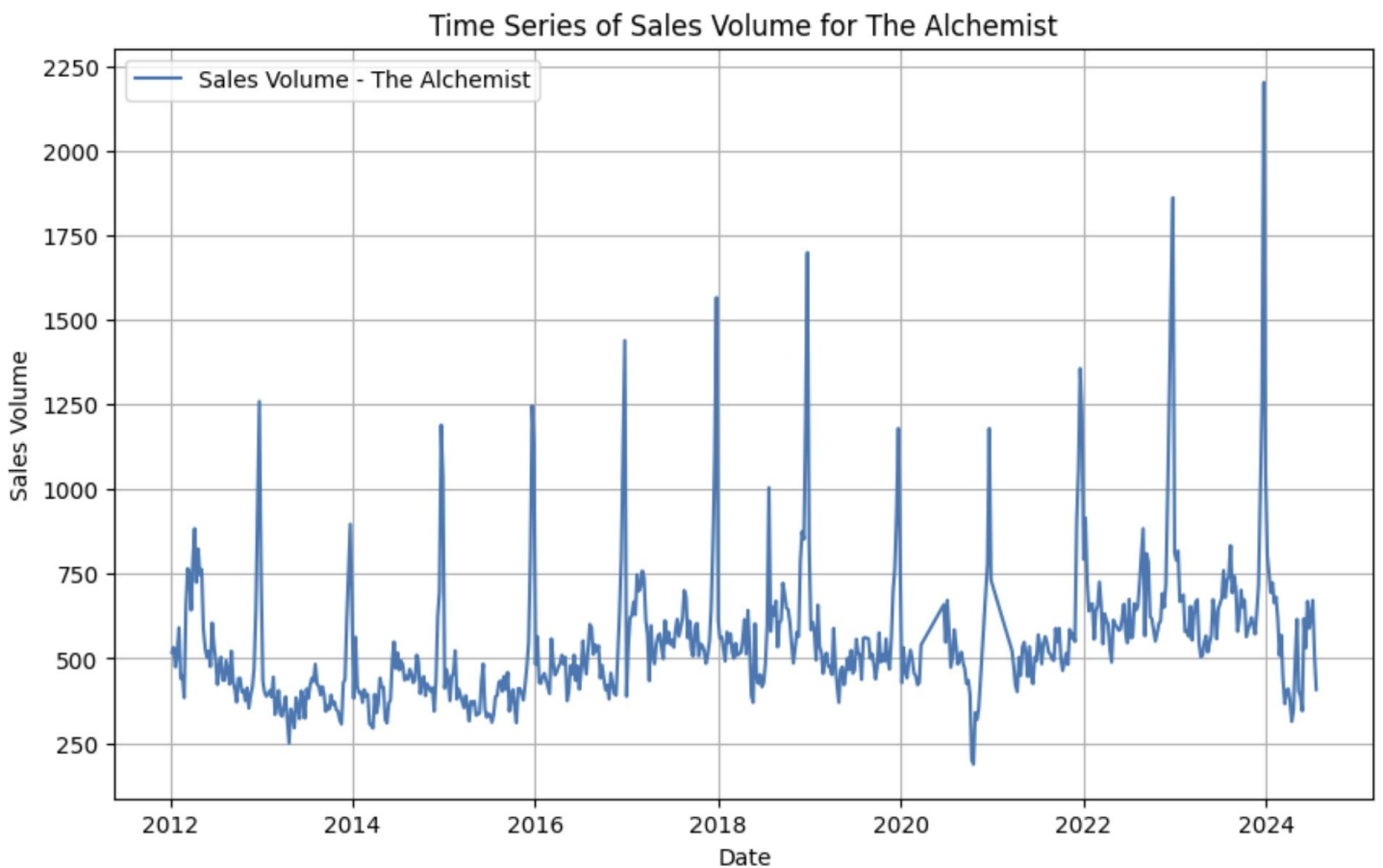
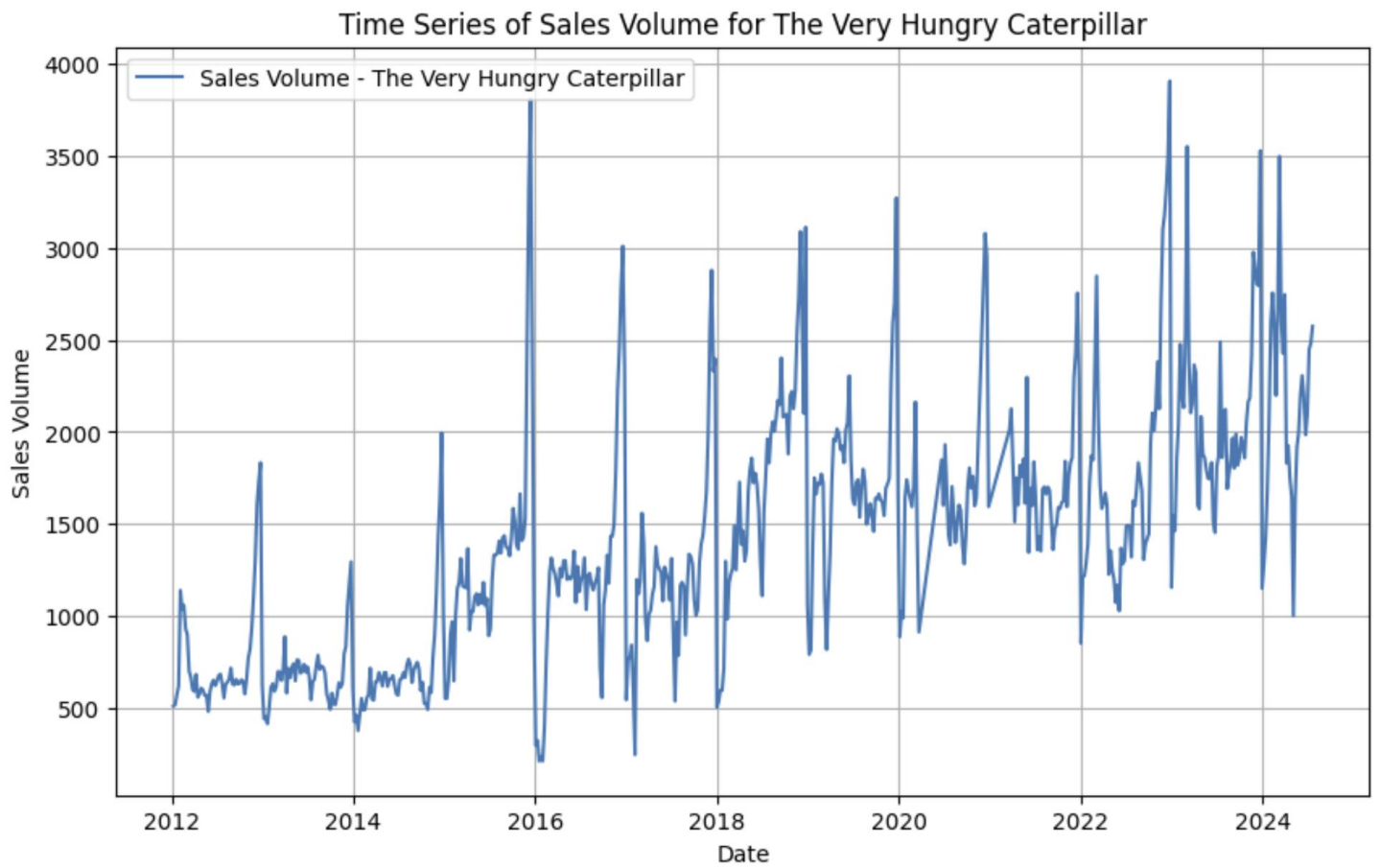


Figure 3: Time series sales volume for The Very Hungry Caterpillar & The Alchemist

3. Interpretation of Time Series Analysis Results

Additive decomposition was carried out on both books (**Figures: Figure 8 & 9**). Both showed strong and consistent seasonal patterns and residuals which hovered close to zero. The trend component of The Very Hungry Caterpillar was observed to be in an upward movement while Alchemist's trend reflected a more volatile pattern.

The ACF and PACF plots for both books (**Figures: Figure 10 & 11**), reflected a slow decay, while the PACF plot identified significant spikes at the first 2 lags, followed by a cut-off or a more rapid decay, with most of the lags falling within the insignificance region.

3.1. Auto ARIMA, XGBOOST, LSTM & Hybrid model Comparison

The Hungry Caterpillar

	MAE	MAPE	Parameters
SARIMA (Classical Model)	353.1416	18.63%	ARIMA(2,1,1)(1,0,2)[52]
XGBoost (Machine Learning model)	539.0560	23.5501%	window_length : 78, learning_rate': 0.05, 'max_depth': 7, 'n_estimators': 500
LSTM (Deep Learning Model)	614.47	34.12%	Best trial: 01, num_lstm_layers: 1, units_lstm_0: 100, dropout_lstm_0: 0.2, optimizer: adam, learning_rate: 0.002411
SARIMA-LSTM Sequential (Hybrid Model)	352.1477	18.5921%	Best trial: 07, num_lstm_layers: 1, units_lstm_0: 100, dropout_lstm_0: 0.5, optimizer: rmsprop, learning_rate: 0.00098
SARIMA-LSTM Parallel (Hybrid Model)	339.9313	17.8647%	Best trial: 09, num_lstm_layers: 1, units_lstm_0: 100, dropout_lstm_0: 0.1, optimizer: rmsprop, learning_rate: 0.00669

The Alchemist

	MAE	MAPE	Parameters
SARIMA (Classical Model)	155.07475	29.75%	ARIMA(1,1,2)(2,0,0)[52]
XGBoost (Machine Learning model)	147.2858	28.7695%	window_length : 12, 'learning_rate': 0.05, 'max_depth': 5, 'n_estimators': 300
LSTM (Deep Learning Model)	258.95	44.98%	Best trial: 07, num_lstm_layers: 2, units_lstm_0: 68, dropout_lstm_0: 0.2, optimizer: adam, learning_rate: 0.0006
SARIMA-LSTM Sequential (Hybrid Model)	164.4254	32.1346%	Best trial: 06, num_lstm_layers: 2, units_lstm_0: 36, dropout_lstm_0: 0.2, optimizer: rmsprop, learning_rate: 0.00160
SARIMA-LSTM Parallel (Hybrid Model)	133.2648	26.6763%	Best trial: 08, num_lstm_layers: 2, units_lstm_0: 100, dropout_lstm_0: 0.2, optimizer: rmsprop, learning_rate: 0.00089

The Hungry Caterpillar (Weekly vs Monthly)

	MAE	MAPE	Parameters
XGBOOST (Weekly)	539.0560	23.5501%	window_length : 78, learning_rate': 0.05, 'max_depth': 7, 'n_estimators': 500
XGBOOST (Monthly)	2369.8550	24.7846%	window_length : 6, 'learning_rate': 0.1, 'max_depth': 5, 'n_estimators': 500
SARIMA (Weekly)	353.1416	18.63%	ARIMA(2,1,1)(1,0,2)[52]
SARIMA (Monthly)	1968.9801	20.0975%	SARIMAX (1, 1, 1)x(1, 0, 1, 12)

The Alchemist (Weekly vs Monthly)

	MAE	MAPE	Parameters
XGBOOST (Weekly)	147.2858	28.7695%	window_length : 12, 'learning_rate': 0.05, 'max_depth': 5, 'n_estimators': 300
XGBOOST (Monthly)	694.9694	34.3244%	window_length : 6, 'learning_rate': 0.1, 'max_depth': 7, 'n_estimators': 500
SARIMA (Weekly)	155.07475	29.75%	ARIMA(1,1,2)(2,0,0)[52]
SARIMA (Monthly)	773.0955	33.2858%	SARIMAX(0, 0, 1)x(1, 0, 1, 12)

Table 2: Classical, ML, Deep Learning and Hybrid model result comparison

The best output/result is highlighted in blue

From the above table (**Table 2**), it can be clearly observed that the parallel hybrid model, SARIMA-LSTM for The Very Hungry Caterpillar achieves the best results with an impressive test **MAE of 339.9313 and MAPE of 17.8647%** with optimal weights **SARIMA=0.7, LSTM=0.3**. The optimal weight distribution identified by the parallel hybrid model reveals that the SARIMA base contributes most of the predictive power (0.7), while the LSTM provides a dampening correction rather than a primary signal (**Figure 4**). The 70/30 split suggests the LSTM's standalone forecast is too volatile or overfit to be trusted at full strength, but its residual patterns contain useful information when blended.

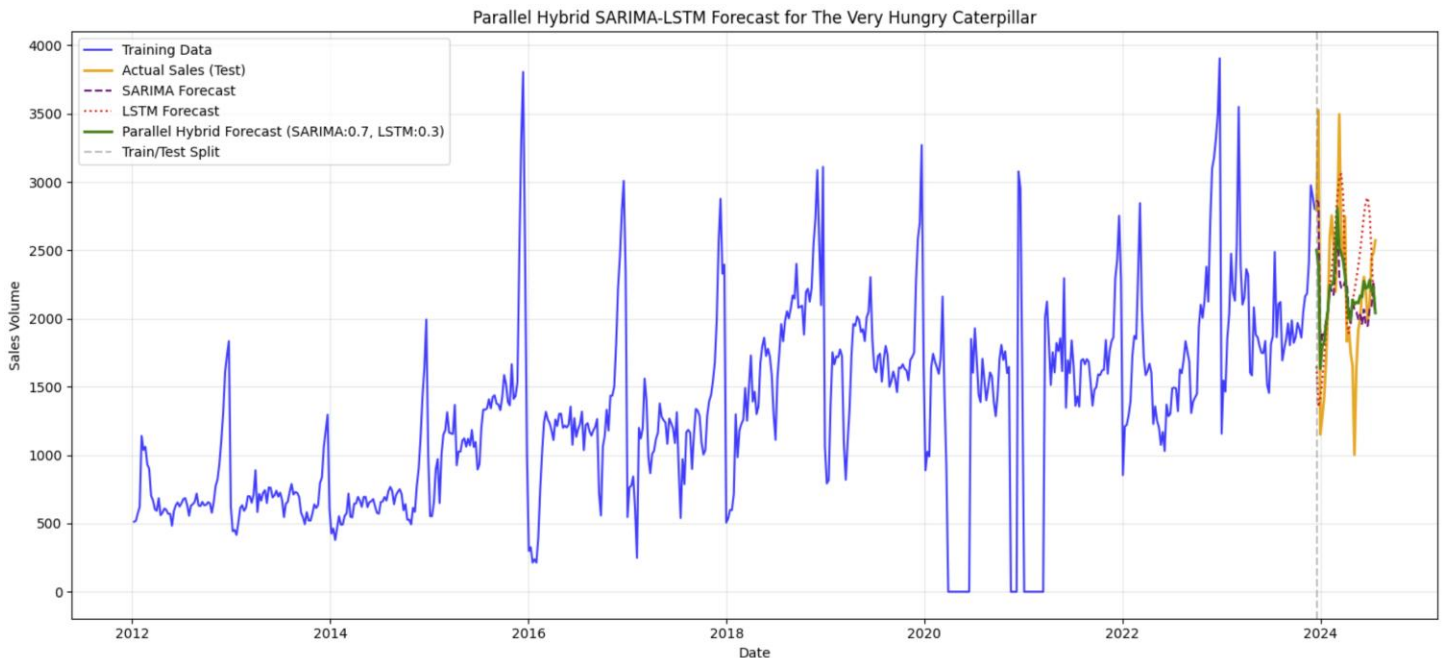


Figure 4: The Very Hungry Caterpillar parallel hybrid SARIMA-LSTM model

Like The Very Hungry Caterpillar model, The Alchemist parallel hybrid SARIMA-LSTM model results in the best performing model for The Alchemist. It achieves a test **MAE of 133.2648 and MAPE of 26.6763%** with equal weights **SARIMA=0.5, LSTM=0.5**. The equal weighting differs from the weighting of The Very Hungry Caterpillar, suggesting the LSTM captures nonlinear patterns that the SARIMA base misses, particularly around the unprecedented test period spike (**Figure 5**). However, the still elevated 26.6763% MAPE is due to the parallel hybrid still underpredicting the initial peak magnitude.

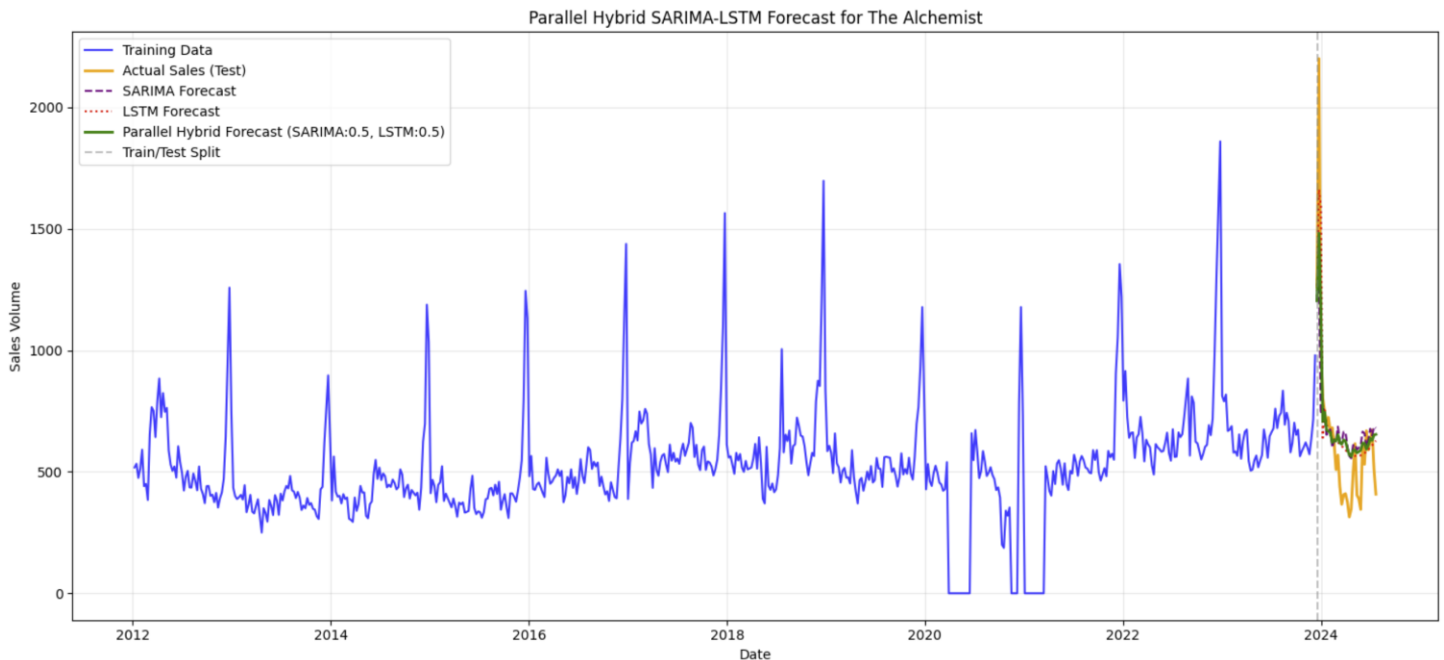


Figure 5: The Alchemist parallel hybrid SARIMA-LSTM model

3.2. Weekly vs Monthly Comparison (XGBoost)

The Very Hungry Caterpillar's XGBoost monthly model's (**Figure 6**) MAE was observed to be much larger than its weekly XGBoost counterpart (**2369.8550 vs 539.0560**). This is not surprising though as; monthly volumes are naturally much larger than weekly volumes. The MAPEs were a better & accurate comparison tool reflecting a nearly identical **24% for both (24.7846% monthly vs 23.5501%)**.

Monthly forecasting usually trades noise reduction against signal loss. For a stable, seasonal series like The Very Hungry Caterpillar, the exchange is fair and results are similar with weekly data thus leaving it unaffected by the trade-off.



Figure 5: The Very Hungry Caterpillar XGBoost (Weekly vs Monthly) model

The Alchemist's XGBoost monthly model's (**Figure 7**) MAPE differed with the weekly MAPE. The weekly MAPE performed better than its monthly counterpart (**34.3244% monthly vs 28.7695%**).

The underperformance of the monthly XGBoost model can be attributed to the inability of the monthly model to capture the volatile and sharp direction of the data. The Alchemist series is more volatile and regime-shifting and thus, the trade-off the monthly model utilizes, is more pronounced as it

discards the very information the model needs to detect anomalies resulting in an underwhelming result.



Figure 7: The Alchemist XGBoost (Weekly vs Monthly) model

4. Conclusions and Recommendations

- For stable seasonal series, hybrid models excel. The parallel SARIMA-LSTM combination for The Very Hungry Caterpillar achieved the lowest error by letting SARIMA capture the structural trend & seasonality (weight = 0.7) while the LSTM corrected residual nonlinearities (weight = 0.3).
- For the volatile, regime-shifting series (The Alchemist), the unprecedented 2024 test spike and 2020 zero sales breaks challenged all models, but the parallel hybrid's error diversification which combined SARIMA's smooth baseline with LSTM's local responsiveness produced the best results.
- Limitations include Recursive multi-step forecasting which caused amplitude damping in the tree and neural models over long periods. The LSTM validation strategy reused training data as validation data, risking overfitting, and the optimal hybrid weights (0.7/0.3 for The Very Hungry Caterpillar, 0.5/0.5 for The Alchemist) are dataset specific and require re-optimization for other books.
- Practically, the weekly parallel hybrid model should be deployed for both stable and volatile book titles, but lower accuracy should be expected for structurally disrupted series. For unpredictable books like The Alchemist, it would be advisable to maintain conservative safety stock buffers regardless of model choice.

5. Next Steps

- Nielsen should apply this understanding to their commission and print volume strategy to be more successful in acquiring titles that have longevity and ensure the correct stock levels. Models and observations noted should be put into consideration when making decisions going forward.
- Findings should be duly conveyed to business stakeholders.

Total word count: 1536 (excluding tables)

6. Figures



Additive Decomposition - The Alchemist



Figure 9: Additive Decomposition (The Alchemist)

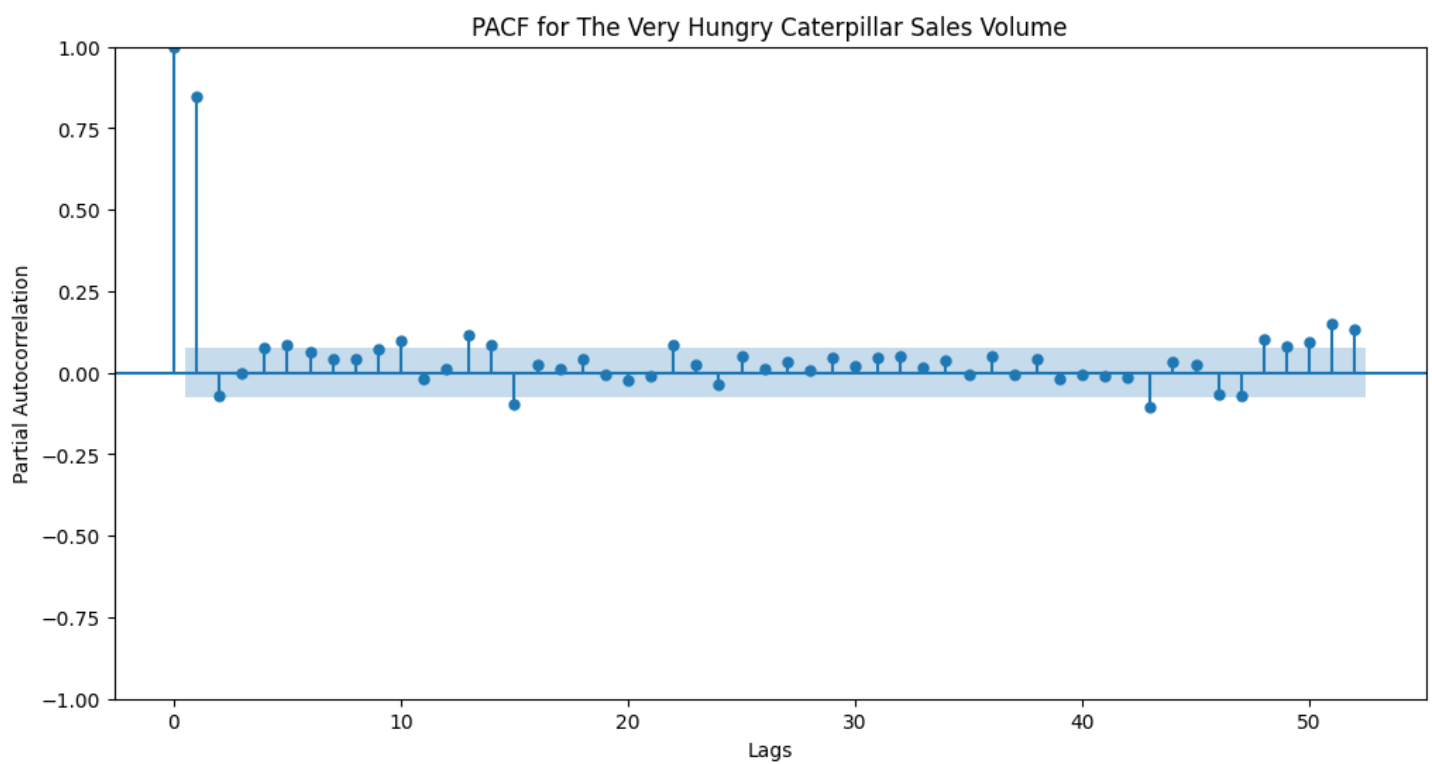
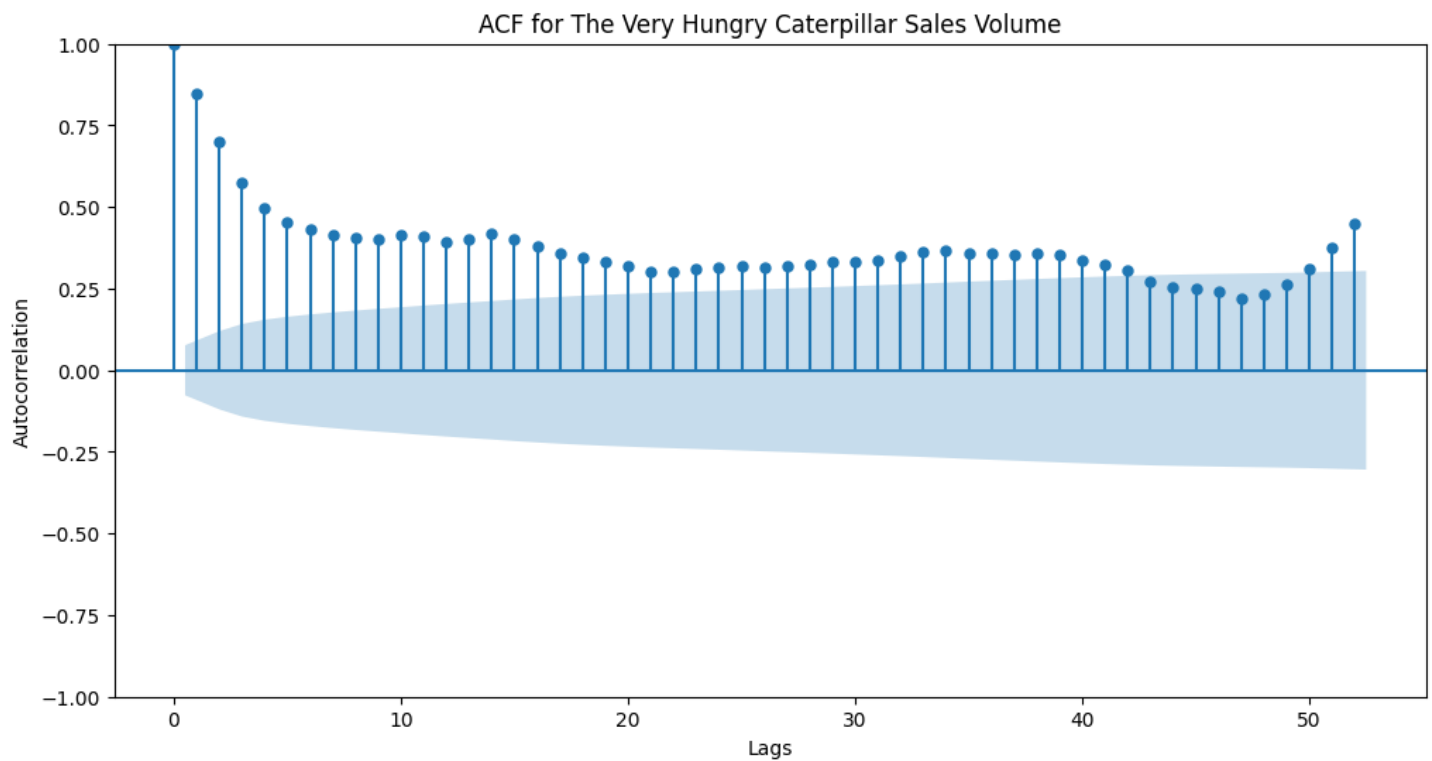


Figure 10: ACF & PACF (The Very Hungry Caterpillar)

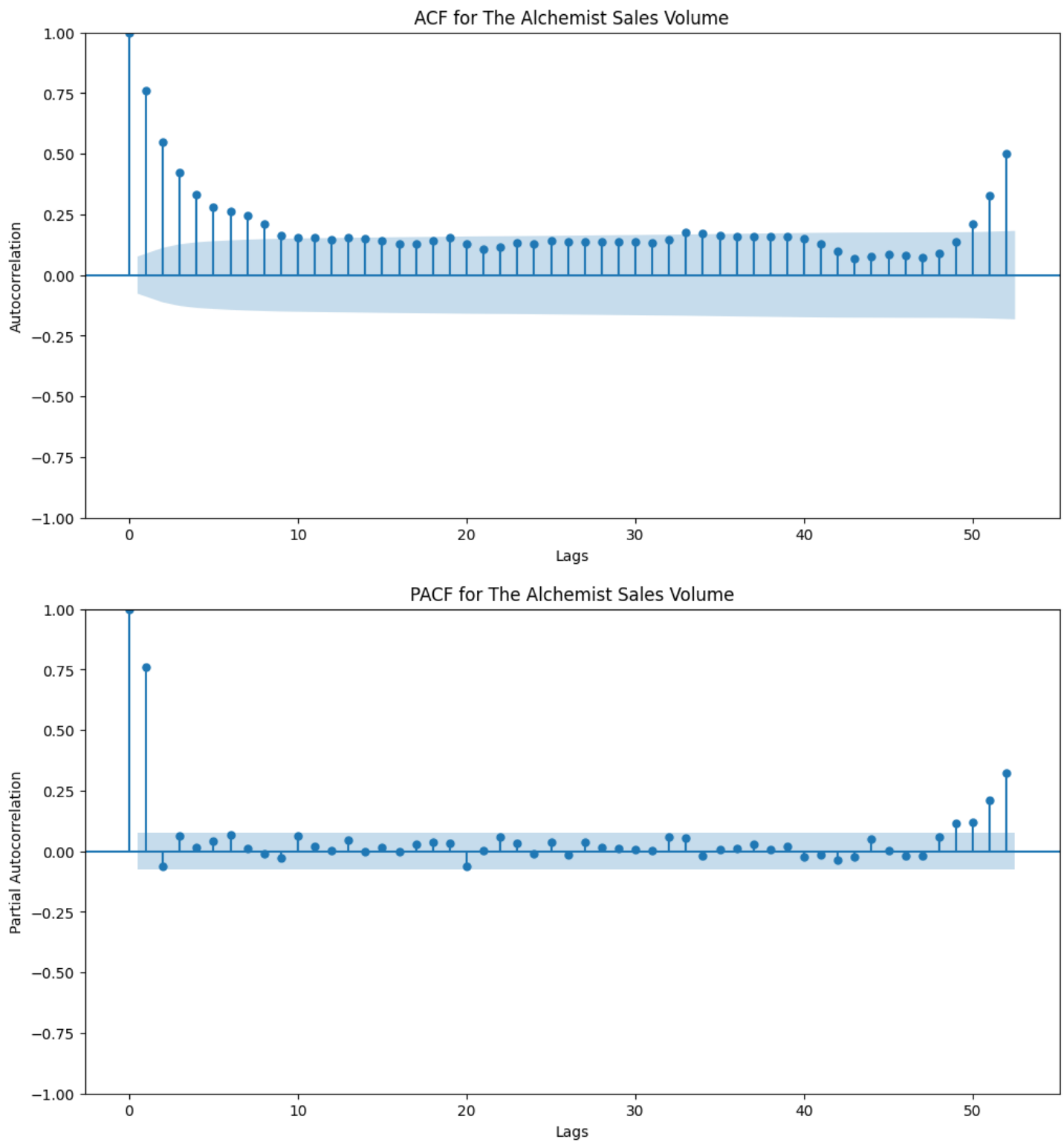


Figure 11: ACF & PACF (The Alchemist)

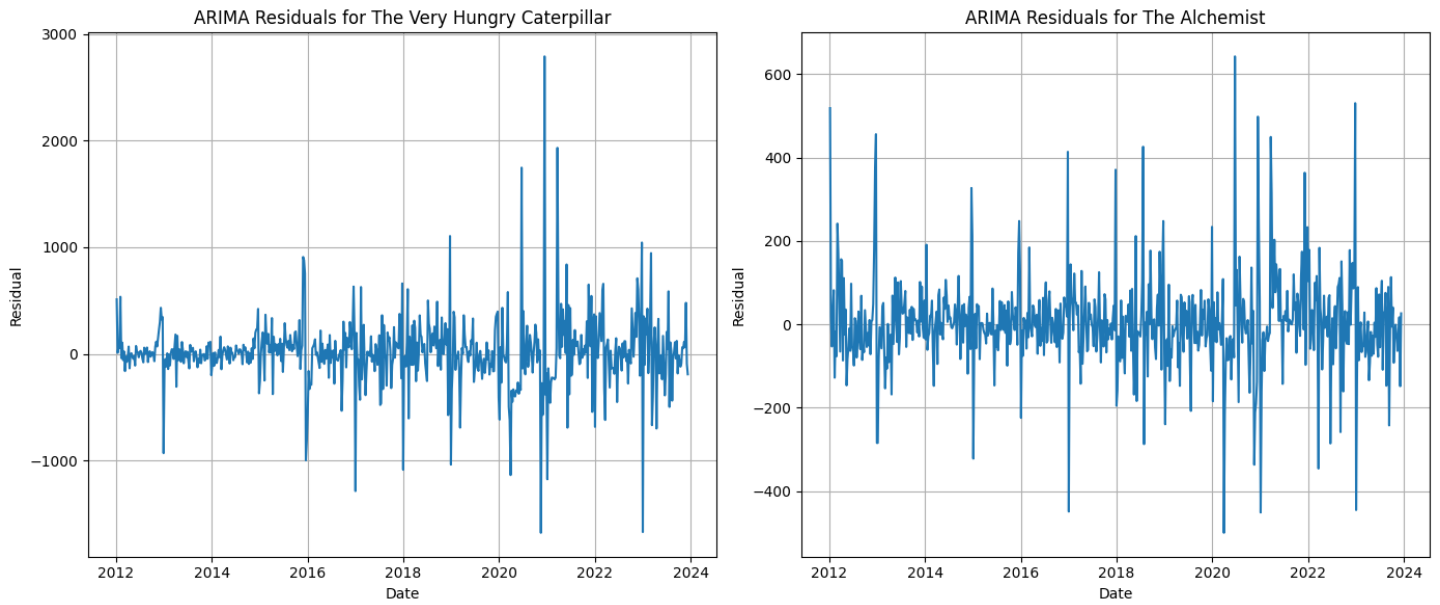
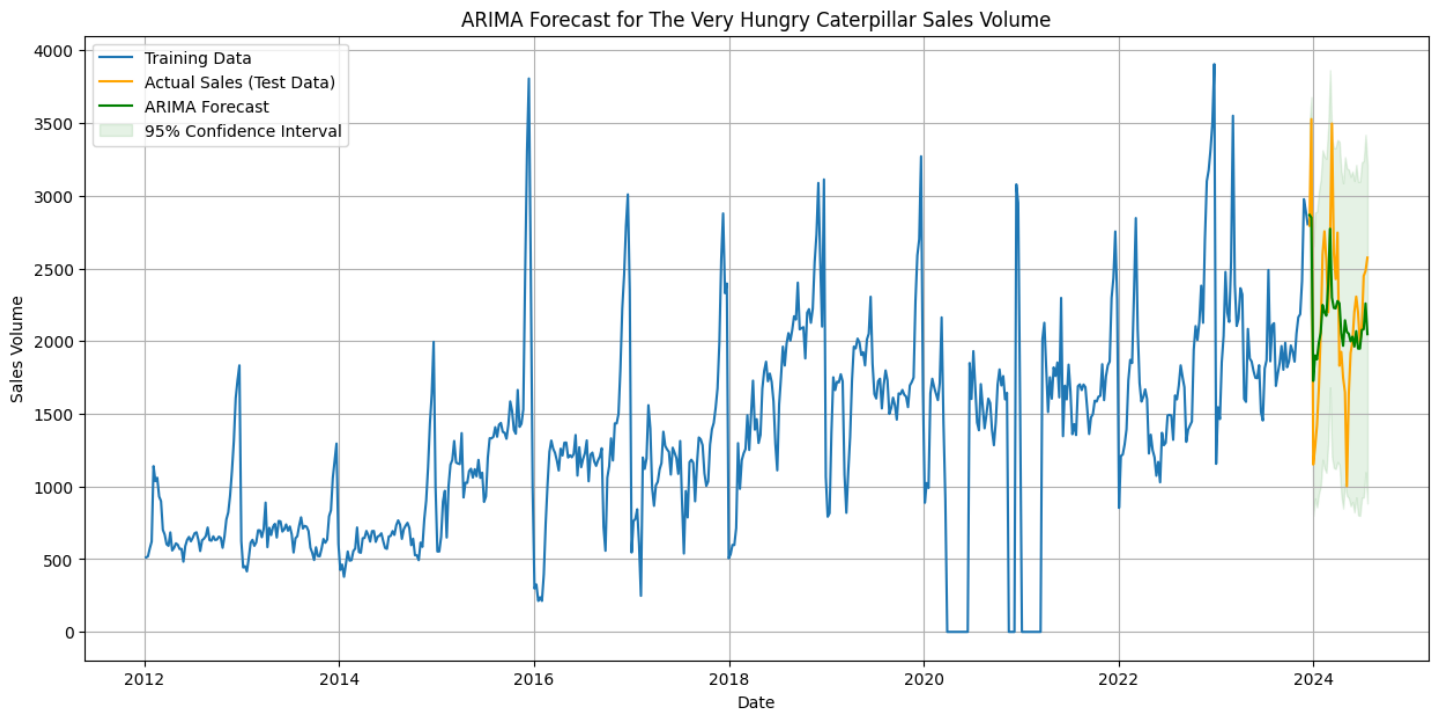


Figure 12: ARIMA Residuals for The Very Hungry Caterpillar & The Alchemist



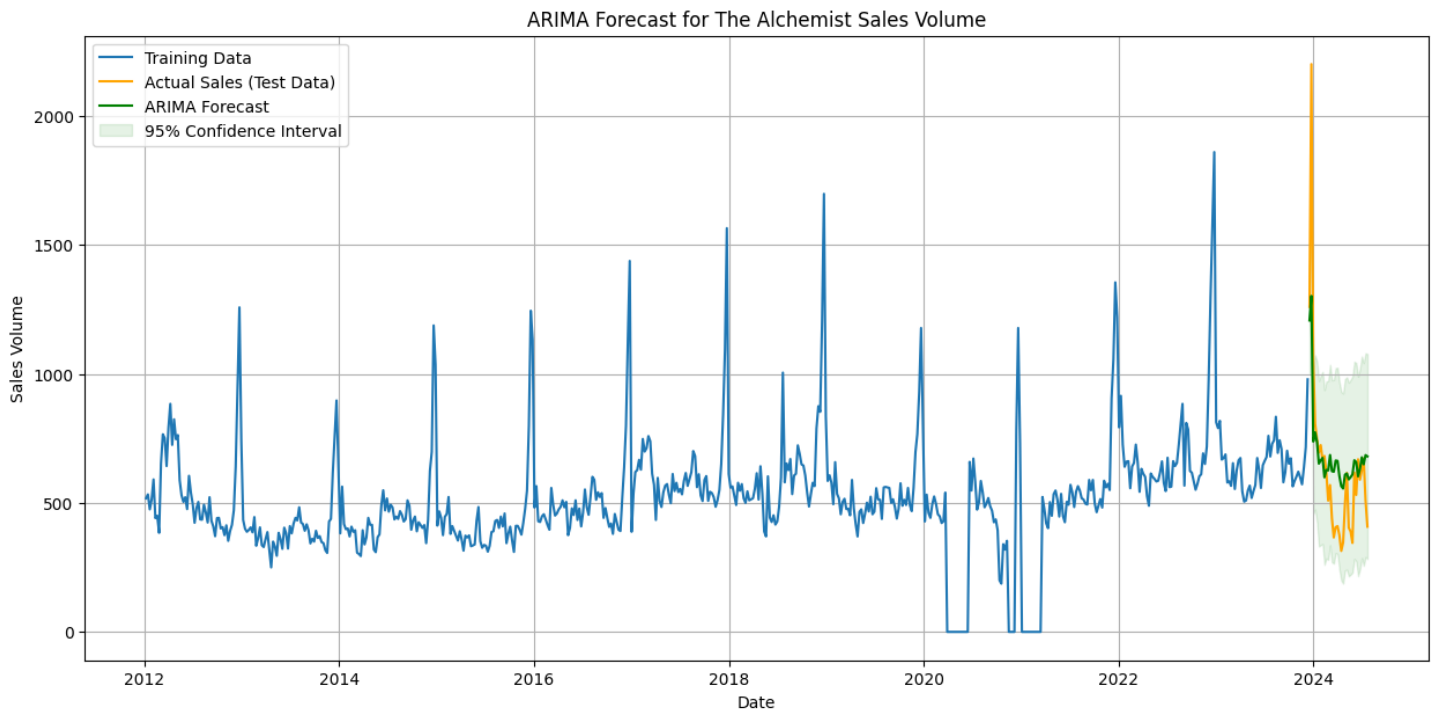
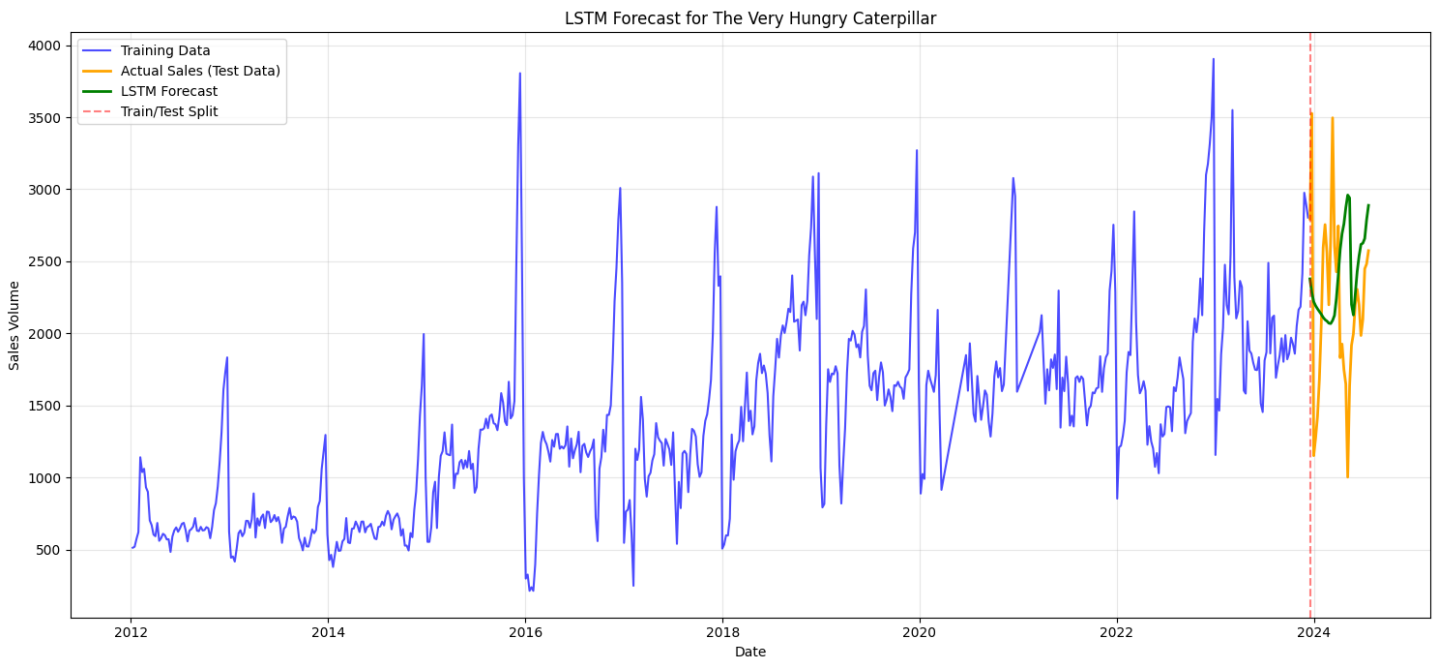


Figure 13: ARIMA forecast for The Very Hungry Caterpillar & The Alchemist



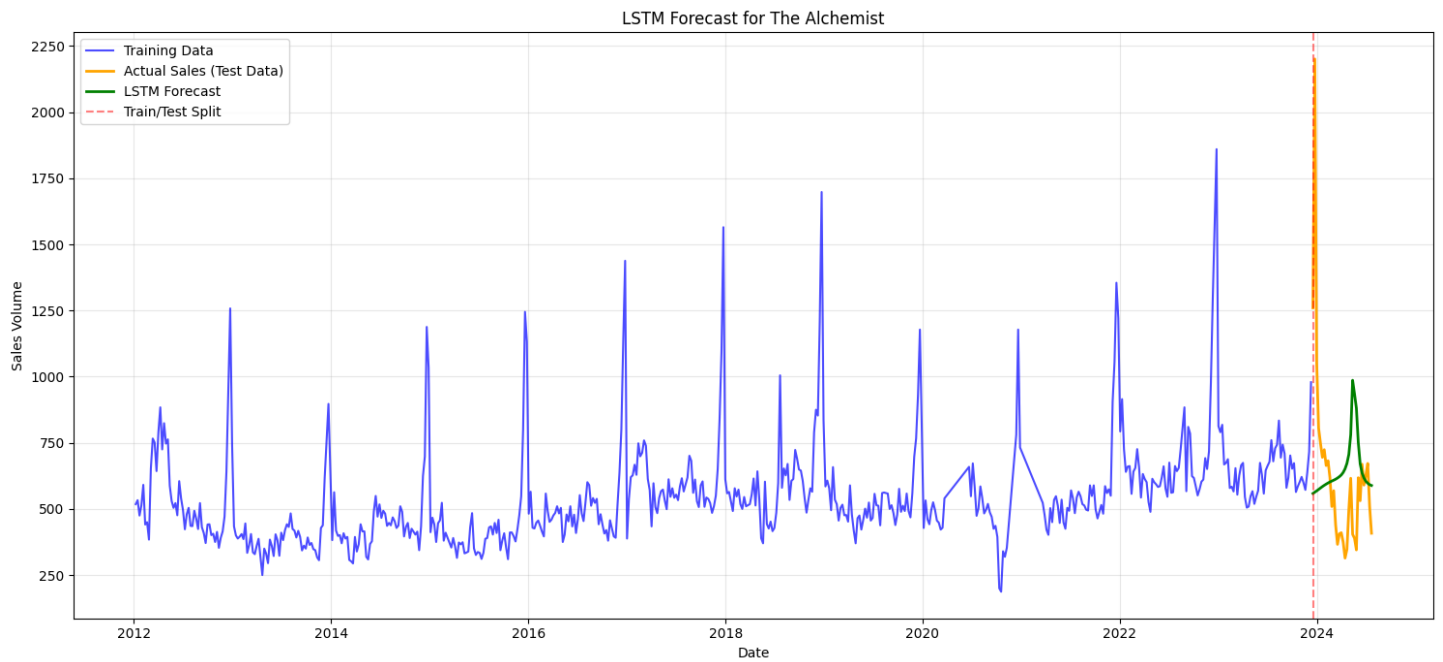
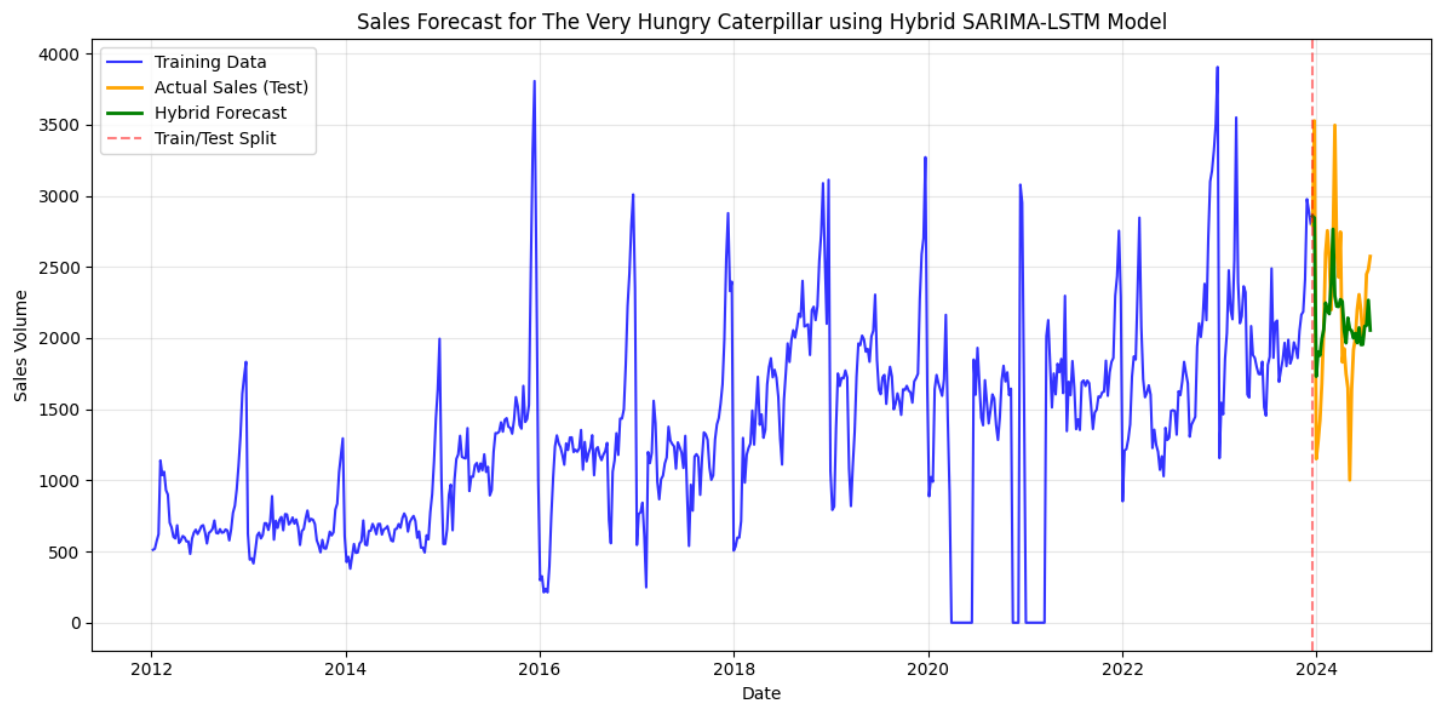


Figure 14: LSTM forecast for The Very Hungry Caterpillar & The Alchemist



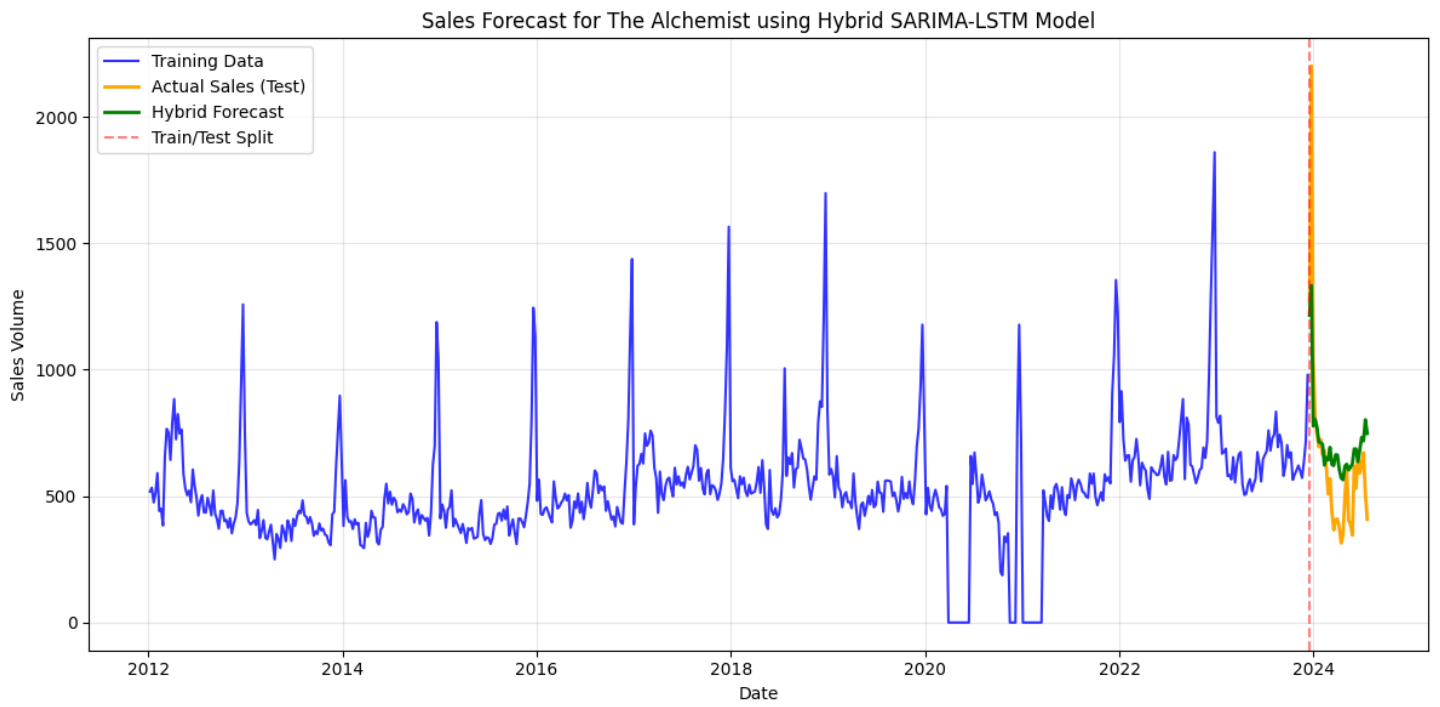
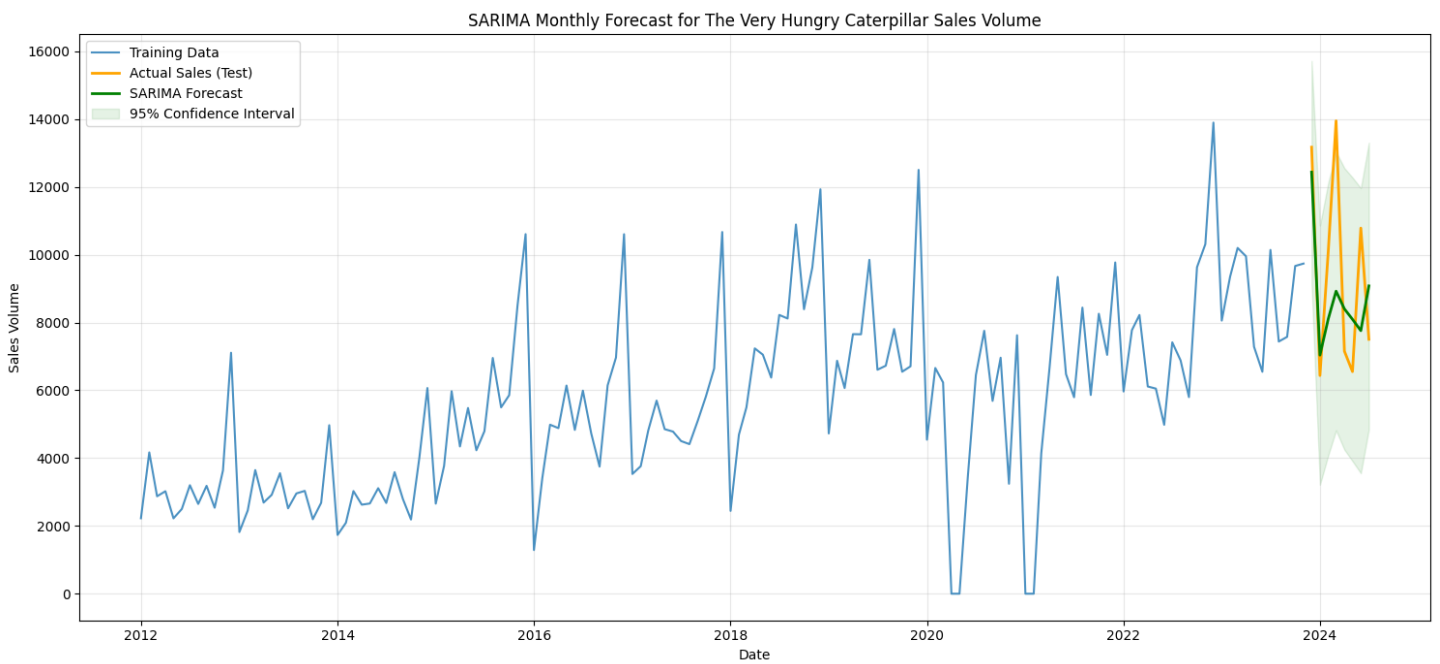


Figure 15: Hybrid SARIMA-LSTM (Sequential) forecast for The Very Hungry Caterpillar & The Alchemist



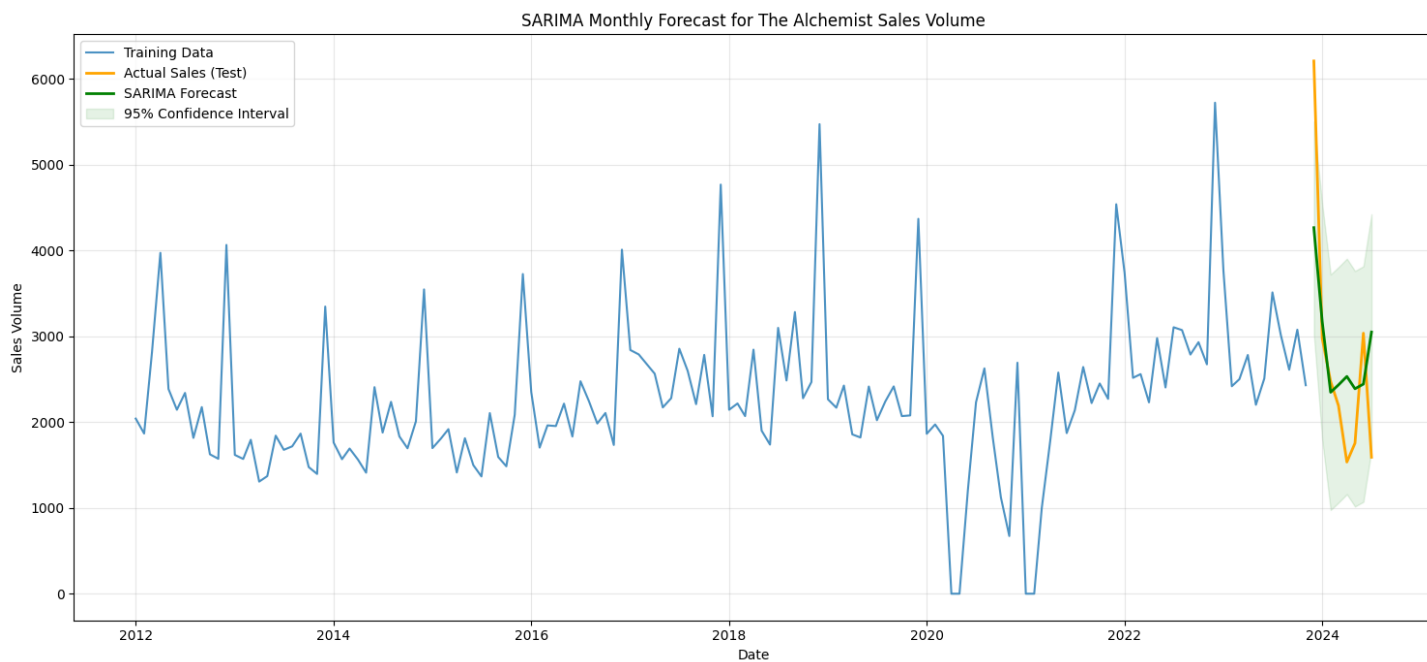


Figure 15: SARIMA monthly forecast for The Very Hungry Caterpillar & The Alchemist